4x4 INT8 Matrix Multiplication Engine

Verification and FPGA Implementation Results

Design `matmul_top` (Verilog) | Target: Xilinx Artix-7 `xc7a35tcpg236-1` | Tool: Vivado 2025.1 | Clock constraint: 10 ns (100 MHz)

205 / 205	+0.354 ns	134 LUTs	104 MHz
tests passed, 0 errors	timing slack after place and route at 10 ns	under 1% of the chip, 0 DSP blocks	estimated maximum clock frequency (Fmax)

This report documents a small digital circuit that multiplies two 4x4 matrices of signed 8-bit numbers. It covers how the circuit works, how it was verified in simulation, and the results of synthesizing and implementing it on an FPGA with Vivado. Every result screenshot is followed by two short notes: a plain-language explanation for readers new to chip design, and the technical detail as reported by the tool.

How to read this report

- **In plain terms** (shaded box): what the screenshot means, with no background knowledge assumed.
- **Technical detail**: exact figures, terminology and caveats, so the result can be checked against the tool output.
- Unfamiliar words are defined in the key terms table in Section 1.

1. Project overview

1.1 The problem: matrix multiplication

A matrix is a grid of numbers. Multiplying two 4x4 grids produces a third 4x4 grid. Each cell of the result takes one row of the first grid and one column of the second: the four pairs are multiplied and the products are added. For example, the row [1, 2, 3, 4] and the column [5, 6, 7, 8] give 1x5 + 2x6 + 3x7 + 4x8 = 70.

A full 4x4 multiplication needs 16 such sums, which is 64 individual multiplications. This operation is central to AI, computer graphics, signal processing and control systems, so processors and accelerators often include dedicated hardware for it. This project is a small educational implementation of that well-known building block. It is not presented as a new design.

1.2 Design approach

The design is written in **Verilog**, a language that describes circuits instead of giving step-by-step instructions. Inputs are signed 8-bit numbers (range -128 to 127). Results are 18 bits wide: the largest single product, (-128) x (-128) = 16,384, needs 16 bits, and adding four such products reaches 65,536, which needs 18 bits.

The architecture uses **one shared multiply-accumulate (MAC) unit** that is reused for all 64 products. Three counters step through every combination: `i` (row of A), `j` (column of B) and `k` (which of the four terms). On each clock cycle the unit multiplies one pair and adds it to a running total. After four terms the finished cell is stored and the total restarts. Sixteen cells at four cycles each gives **64 clock cycles per full multiplication**. Three small memories hold matrix A (16 x 8 bits), matrix B (16 x 8 bits) and the results (16 x 18 bits).

The trade-off is deliberate: a single multiplier keeps the hardware small, at the cost of speed compared with a fully parallel design (which was not built).

Data enters and leaves through a compact 40-pin interface: load 32 values, pulse `start`, wait for `done`, then read the 16 results.

Pin	Direction	Width	Function
`clk`	in	1	Clock that sequences all operations
`rst`	in	1	Reset (active high), clears the control logic
`wr_en`	in	1	Write enable: store `wr_data` on this clock tick
`wr_sel`	in	1	Target matrix: 0 = A, 1 = B
`wr_addr`	in	4	Element index 0 to 15 (row x 4 + column)
`wr_data`	in	8	Signed value to store (-128 to 127)
`start`	in	1	Pulse high for one clock to begin the calculation
`done`	out	1	High when the results are ready
`rd_addr`	in	4	Which result element to read
`rd_data`	out	18	Signed 18-bit result element

Total: 40 pins (21 inputs including the clock, 19 outputs).

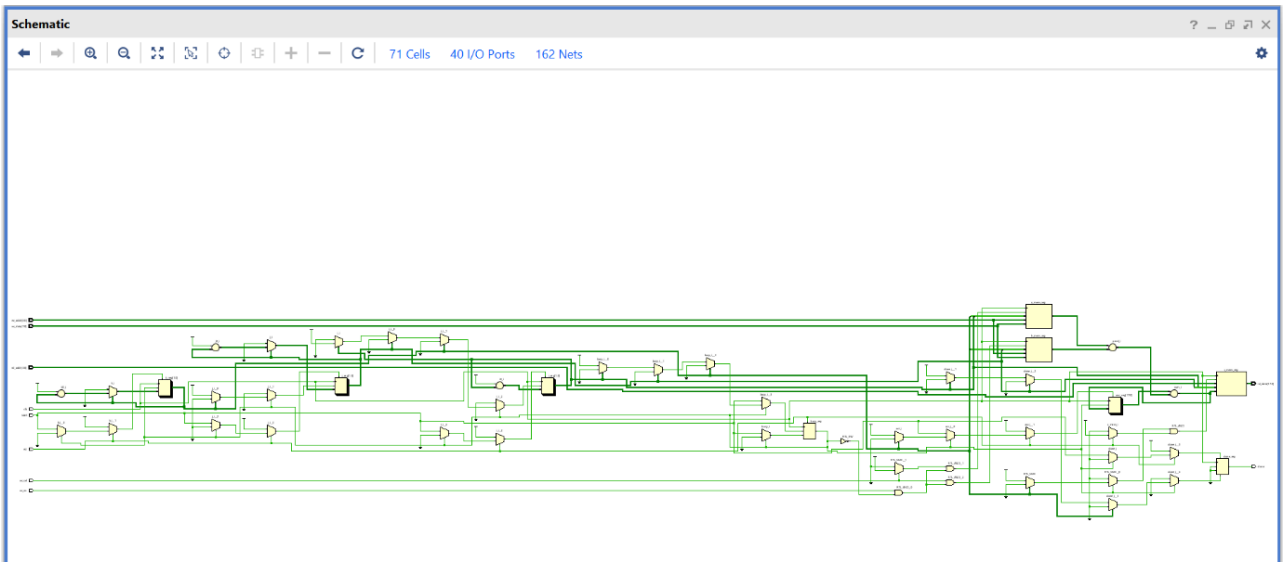


Figure 1. *RTL schematic of matmul_top (Vivado elaborated design).*

> **In plain terms.** This is Vivado's drawing of what the Verilog code describes, before any optimization: blocks that select, add, multiply and store values, joined by wires. It works like a blueprint of the circuit. The 40 ports on the diagram match the 40 pins in the table above.

Technical detail. Elaborated (RTL) netlist with 71 cells, 40 I/O ports and 162 nets. A cell is a building block and a net is a wire connecting cells. At this stage the blocks are generic (multiplexers, adders, registers) and are not yet mapped to the physical resources of the FPGA.

1.3 Key terms

Term	Meaning
Verilog	A hardware description language: it describes the structure and behavior of a digital circuit.

Term	Meaning
RTL	Register-transfer level: a circuit described as registers and the logic between them.
Testbench	A file used only in simulation. It feeds inputs to the design and checks the outputs.
Behavioral simulation	Running the Verilog on a computer to check that it works, without physical timing delays.
FPGA	A chip whose logic can be reprogrammed after manufacture.
LUT	Look-up table: the small programmable logic element that acts as the FPGA's basic gate.
Register (flip-flop)	A one-bit memory that captures its input on each clock tick.
DSP block	Dedicated multiplier and adder hardware found on many FPGAs.
Clock period	Time between clock ticks. 10 ns corresponds to 100 MHz.
Synthesis	Translating the Verilog into the building blocks of the target chip.
Implementation	Placing those blocks at physical locations on the chip and routing the wires (place and route).
Setup slack (WNS)	Spare time on the slowest path. Positive means every signal arrives before the next clock tick.
Hold slack (WHS)	The opposite check: data must not change too soon after a clock tick. Positive means it passes.
TNS	Total negative slack: the sum of all timing failures. 0 means none.
Fmax	Highest clock frequency at which timing is still met, here estimated as 1 / (period - WNS).
Static / dynamic power	Static: consumed just by being powered. Dynamic: extra consumed when signals switch.

2. Results at a glance

All values below come from Vivado post-implementation reports for the `xc7a35tcpg236-1` device at a 10 ns clock constraint (shown in detail in Section 4).

Metric	Value	What it means
Slice LUTs	134 (110 as logic, 24 as memory)	Logic elements used: 0.64% of the 20,800 available.
Slice registers	26	Flip-flops used: the counters, two state flags and the 18-bit running total.
Slices	41	Groups of LUTs and registers: 0.50% of the 8,150 available.
DSP blocks	0	No dedicated multiplier blocks were used. The multiplication was built from general-purpose logic.
I/O pins	40 of 106	Matches the 40-pin interface.
Clock period constraint	10 ns	Target of 100 MHz.

Metric	Value	What it means
WNS (setup slack)	+0.354 ns	The slowest path finishes 0.354 ns before its deadline. Positive means timing is met.
WHS (hold slack)	+0.097 ns	Hold check passes.
Failing endpoints	0	No path misses its deadline.
Shortest achievable period	10 - 0.354 = 9.646 ns	Estimated from the remaining slack.
Fmax estimate	1 / 9.646 ns, about 104 MHz	Estimated maximum clock frequency.
Cycles per 4x4 multiply	64	Fixed by the architecture: 16 cells x 4 terms.
Time per multiply	64 x 9.646 ns, about 617 ns	Time for one complete 4x4 multiplication at the estimated maximum clock.
Total on-chip power	0.089 W (estimate)	Rough estimate, low confidence (see Figure 12).

Reading the numbers

- Timing is met after real placement and routing, with a small margin. Post-synthesis slack was +0.379 ns, so implementation used about 0.025 ns, which is normal.
- The LUT count fell from 140 after synthesis to 134 after implementation, consistent with further netlist optimization during implementation.
- The 24 memory LUTs hold the three small memories as distributed RAM. This is also why only 26 registers appear: 6 counter bits, 2 state flags (`busy`, `done`) and the 18-bit accumulator.
- Of the 0.089 W power estimate, 0.070 W is static (the chip's own leakage regardless of the design) and about 0.018 W is dynamic, most of it (0.016 W) in the I/O pins. Vivado marks the confidence as low because no real switching activity was supplied, so the figure is a rough estimate and is not comparable with results from other platforms.

Note on results. Figures are from Vivado post-implementation reports for the xc7a35tcpg236-1 device at a 10 ns clock constraint. Timing was met with +0.354 ns setup slack. Memories were mapped to distributed LUT RAM. Power is a vectorless estimate with low confidence.

3. Verification evidence

3.1 How the design was tested

A testbench loads two matrices into the design, starts the calculation, reads all 16 results and compares every cell with its own independent calculation written as plain loops. A mismatch is printed and counted. In total, 205 tests were run: five directed tests chosen for specific reasons, then 200 random matrix pairs.

Test	Inputs	Why it is included
1	All zeros	Baseline: every output must be 0.
2	A = 1 to 16, B = identity matrix	The result must equal A, which is easy to verify by eye.
3	All -128 x all -128	Largest magnitude (65,536 per cell): checks that 18 bits are enough.
4	All +127 x all +127	Largest positive-only case (64,516 per cell).
5	All -128 x all +127	Mixed signs: every result is -65,024.
6 to 205	200 random matrix pairs	Broad coverage of values the directed tests do not reach.

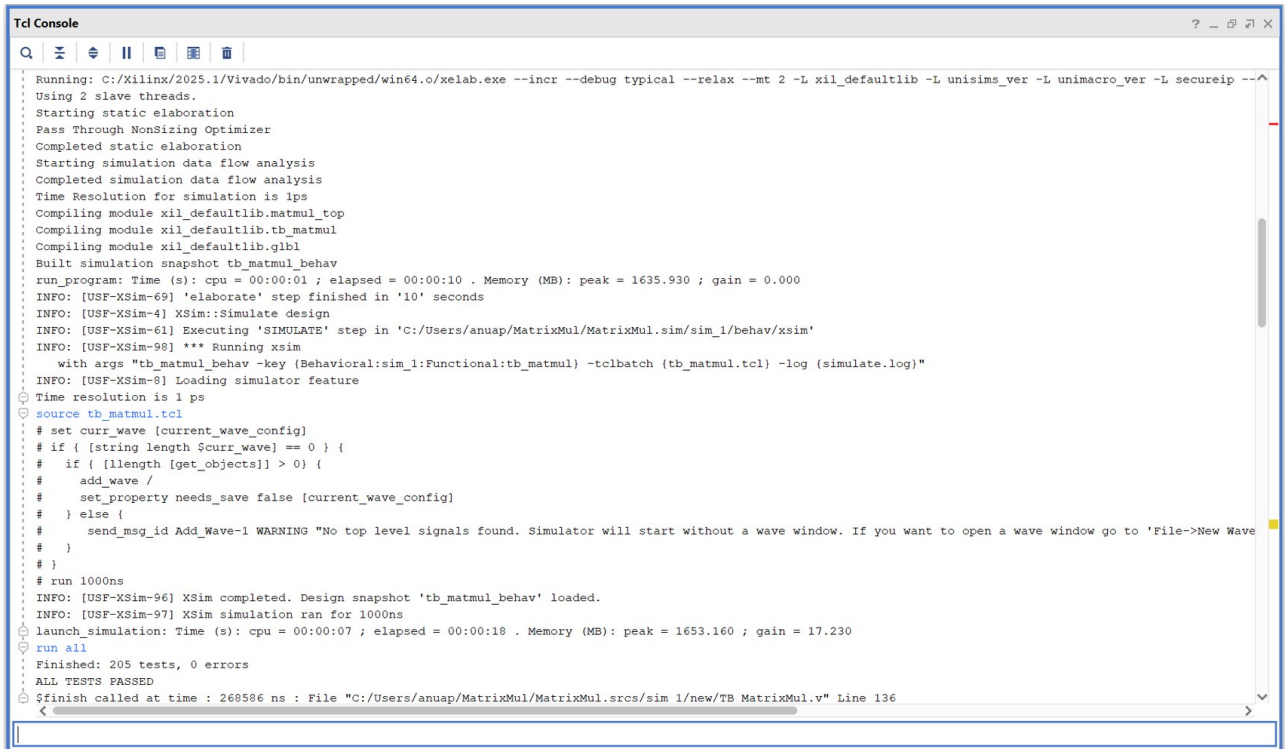


Figure 2. *Simulation console after the full run: all tests passed.*

> **In plain terms.** The simulator runs every test automatically. For each one, all 16 answers are compared with an independent calculation. The last lines report that 205 tests ran, with 0 errors, and print ALL TESTS PASSED.

Technical detail. Vivado XSim behavioral simulation, started with `run all`. The line `Finished: 205 tests, 0 errors` is printed by the testbench, and `$finish` is reached at 268,586 ns of simulated time (about 268.6 microseconds, which took seconds of real computer time). Behavioral simulation checks function only: it does not include physical delays, which are covered by the timing reports in Section 4.

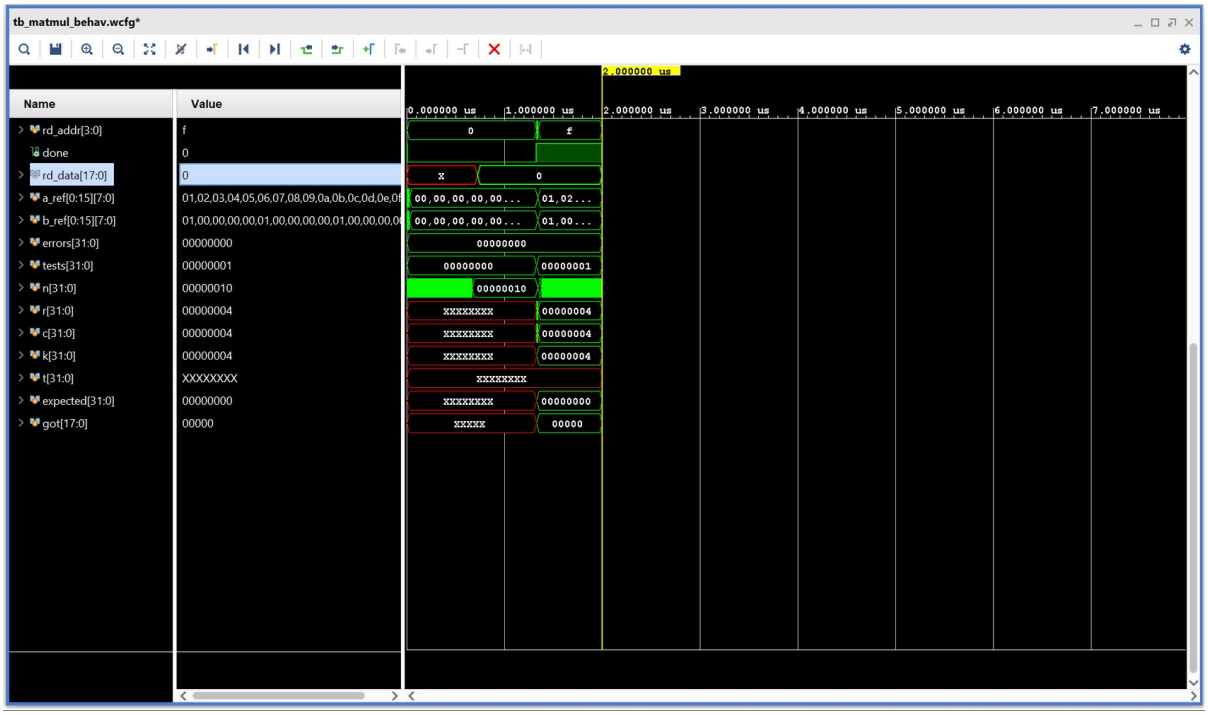


Figure 3. *Waveform of the first test (all-zero matrices), cursor at 2.0 µs.*

> **In plain terms.** A waveform works like an oscilloscope trace: each row is one signal and the horizontal axis is time. The `done` signal rises at about 1.3 µs, which is the moment the chip announces that the answer is ready. `rd_addr` then steps from 0 to f (15 in hexadecimal) as the testbench reads the 16 results. `rd_data` shows 0 because all-zero inputs produce all-zero outputs. The red "x" at the start means "not yet defined" because the result memory has not been written yet. This is expected and is not an error. The `errors` row stays at zero and `tests` is 1.

Technical detail. The timing matches the design: 32 writes at 2 clock cycles each (10 ns clock) take 640 ns, followed by 64 clock cycles (640 ns) of calculation, so `done` rises at about 1.3 µs. At the cursor, the testbench has started loading test 2 (`a_ref` begins 01, 02, ... and `b_ref` begins 01, 00, 00, ... for the identity matrix). Values are displayed in hexadecimal by default.

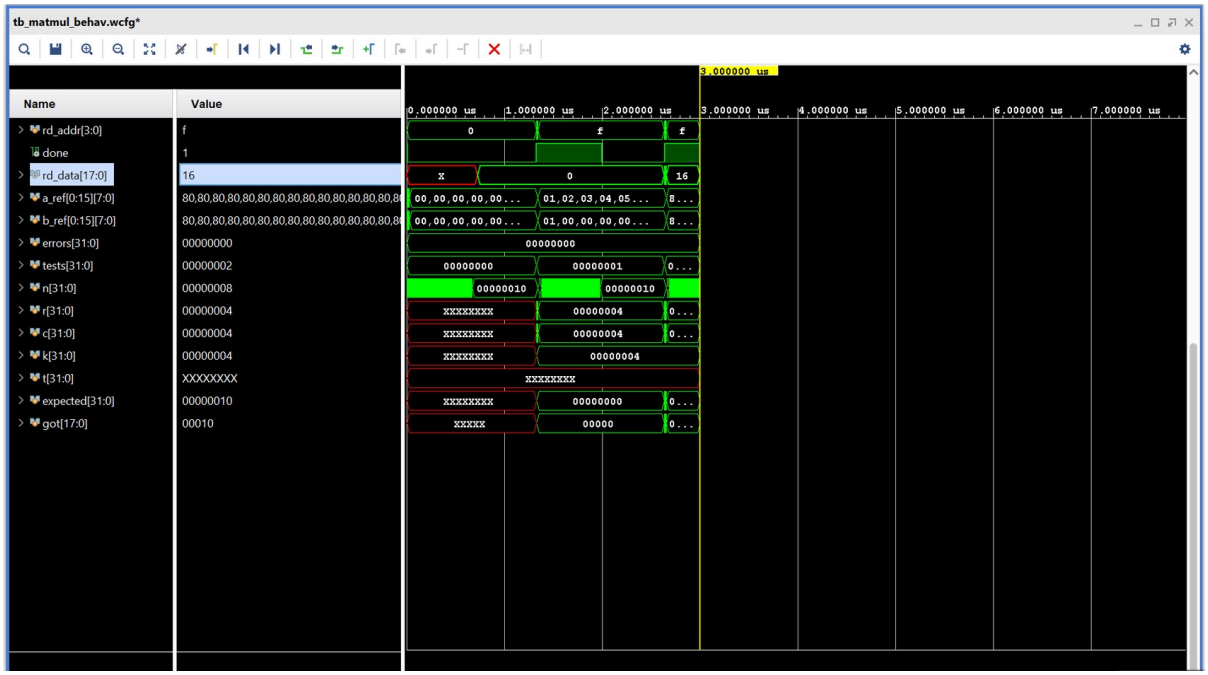


Figure 4. Waveform at 3.0 µs, after the second test (A = 1 to 16 times the identity matrix).

In plain terms. The second test multiplies the grid 1 to 16 by the identity grid (ones on the diagonal, zeros elsewhere). The answer must be identical to the first grid. Reading the last result cell gives 16, exactly the last number of the first grid. `done` fell when the second calculation started at about 2.0 µs and rose again at about 2.6 µs. The counters now show `tests` = 2 and `errors` = 0.

Technical detail. `rd_data` = 16 at `rd_addr` = f (element 15). In the lower rows, `expected` = 0x10 and `got` = 0x10 (both 16 in decimal, shown in hexadecimal). At the cursor, `a_ref` and `b_ref` show 80 (hexadecimal), which is -128: the testbench is already loading test 3.

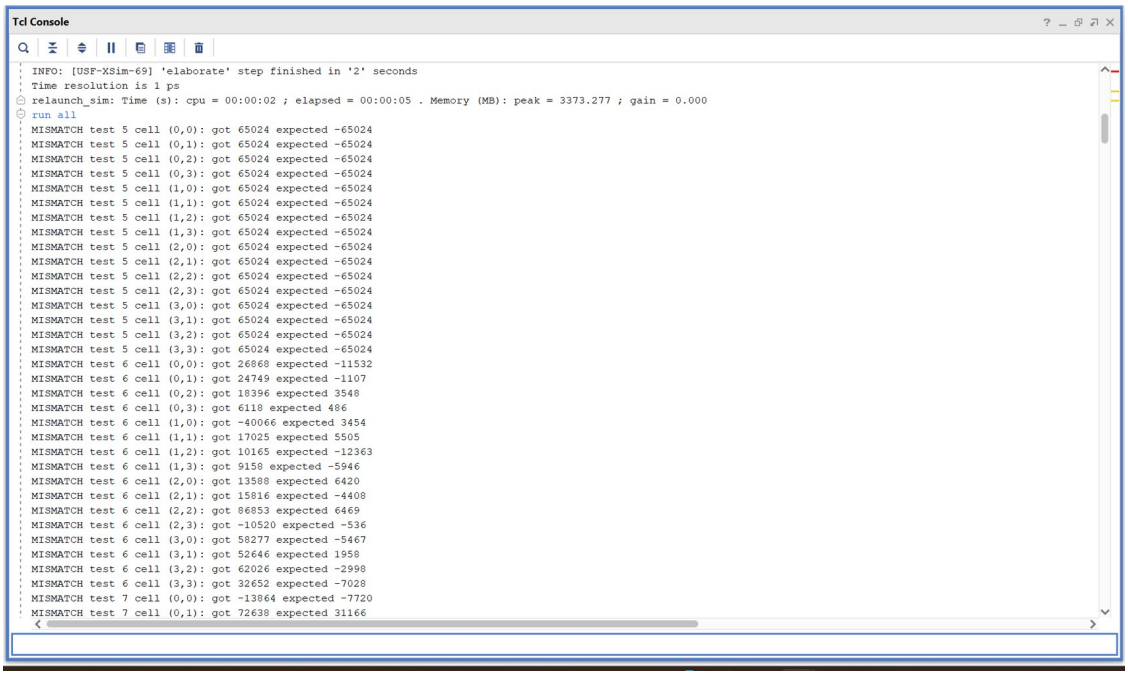


Figure 5. Deliberate fault injection: the testbench reports mismatches (Tcl console).

In plain terms. To prove that the tests can actually catch mistakes, a bug was inserted on purpose: the word `signed` was removed from one signal, so the circuit treated negative numbers as large positive ones. The testbench immediately lists every wrong answer. A test that has never been seen to fail cannot be trusted, so this step supports the credibility of the passing result in Figure 2.

Technical detail. With the fault present, tests 1 to 4 produced no mismatches, because zero, positive-only and same-sign cases still give correct results. The first failure appears at test 5, where every cell reads `got 65024 expected -65024`: the value -128 (binary 10000000) was interpreted as +128. Random tests then fail broadly (for example, test 6, cell (0,0): `got 26868 expected -11532`). This shows why a mixed-sign directed test and random testing are both needed. The fault was removed before synthesis.

4. FPGA synthesis and implementation results

4.1 The two stages

Synthesis translates the Verilog into the building blocks of the target chip (LUTs, registers, memories) and estimates timing. **Implementation** then places each block at a physical location and routes the wires, which gives more realistic timing. Both stages used the same one-line constraint file, `create_clock -period 10.000 -name clk [get_ports clk]`, which sets a 100 MHz target clock.

4.2 After synthesis

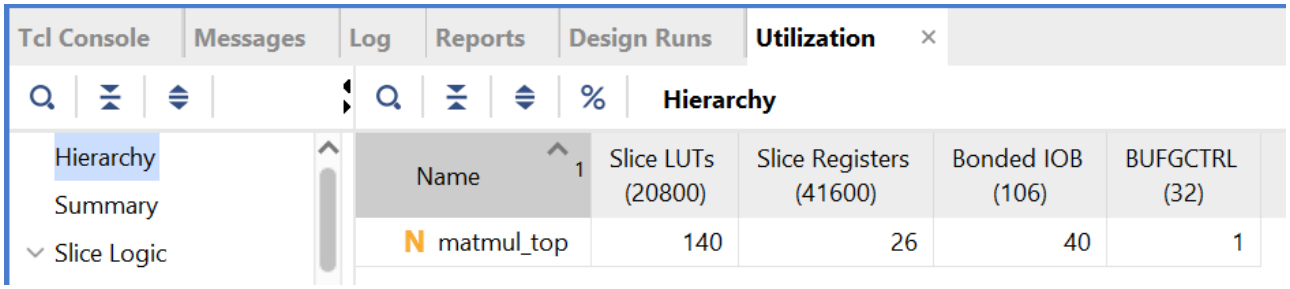


Figure 6. *Post-synthesis utilization (Vivado Utilization report).*

> **In plain terms.** Utilization shows how much of the chip the design consumes. Here it needs 140 of the 20,800 logic elements (LUTs), 26 of 41,600 registers, 40 of 106 pins and one clock buffer. The design uses well under 1% of the chip's logic, so it is very small.

Technical detail. Slice LUTs 140 / 20,800 (0.67%); slice registers 26 / 41,600 (0.06%); bonded IOB 40 / 106 (37.7%); BUFGCTRL 1 / 32. The 26 registers are 6 counter bits (`i`, `j`, k), `busy`, `done` and the 18-bit accumulator. The three memories were mapped to distributed (LUT) RAM, so they do not appear as registers.

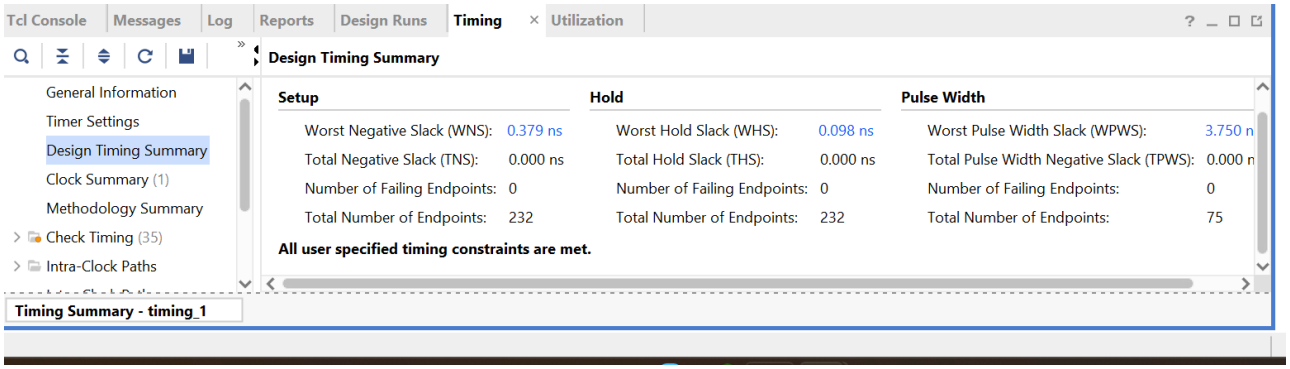
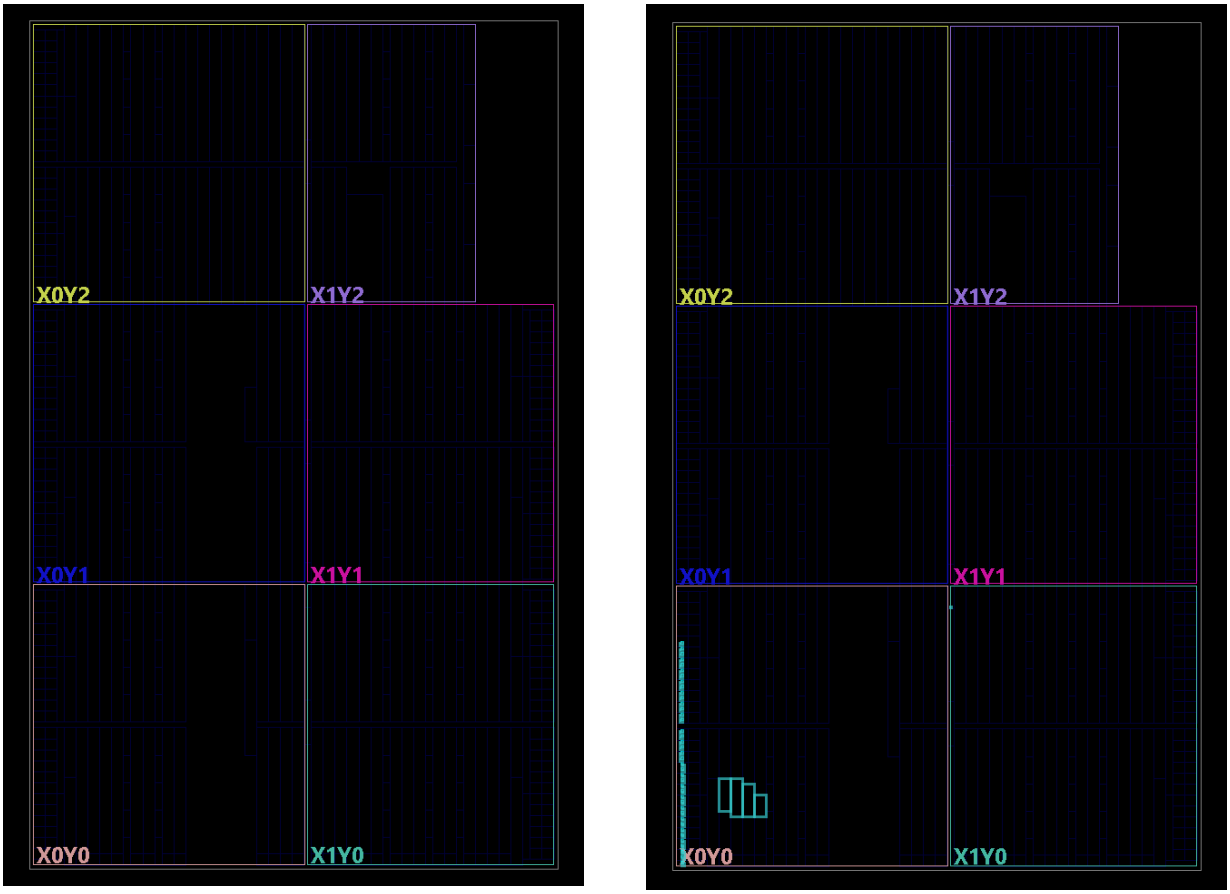


Figure 7. *Post-synthesis Design Timing Summary.*

> **In plain terms.** Every signal path must finish within one clock tick, which is 10 ns here. The worst negative slack (WNS) is the time the slowest path has left over. At +0.379 ns it is positive, so every path arrives on time, and Vivado states that all timing constraints are met. The hold check, which makes sure data does not change too soon after a tick, also passes.

Technical detail. Setup: WNS +0.379 ns, TNS 0.000 ns, 0 failing of 232 endpoints. Hold: WHS +0.098 ns, THS 0.000 ns, 0 failing of 232 endpoints. Pulse width: WPWS 3.750 ns, 0 failing of 75 endpoints. Post-synthesis timing uses estimated wire delays and is therefore optimistic until implementation.

4.3 After implementation



(a) After synthesis: nothing placed yet

(b) After implementation: design placed

Figure 8. *Device view of the xc7a35t, divided into clock regions X0Y0 to X1Y2.*

In plain terms. This is a map of the chip, divided into six zones called clock regions. After synthesis the map is empty, because no block has a location yet. After implementation, the design appears as a small highlighted cluster in the lower-left region. The whole design occupies only a tiny part of the chip.

Technical detail. Implementation (placement and routing) assigned each used resource to a physical site. The placed design is concentrated in region X0Y0, consistent with the under-1% utilization.

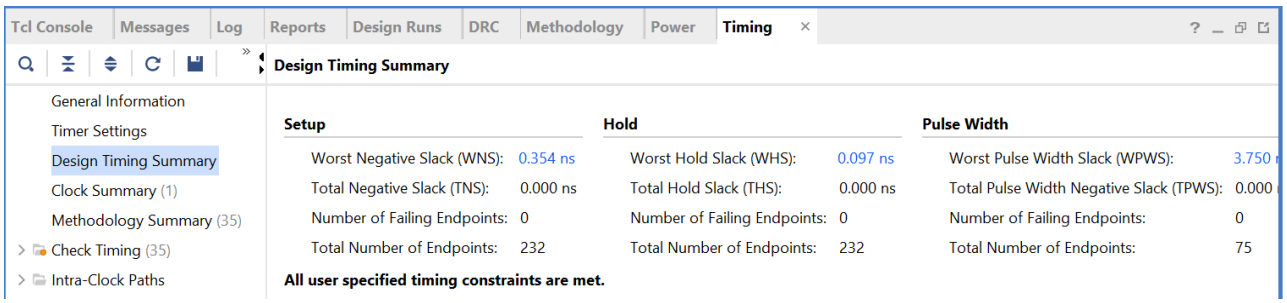


Figure 9. *Post-implementation Design Timing Summary.*

In plain terms. The same checks were repeated after real placement and routing. The slack dropped slightly from +0.379 ns to +0.354 ns but is still positive, so the circuit still meets the 10 ns target. A small margin is normal at this stage.

Technical detail. Setup: WNS +0.354 ns, TNS 0.000 ns, 0 failing of 232 endpoints. Hold: WHS +0.097 ns, THS 0.000 ns, 0 failing of 232 endpoints. Pulse width: WPWS 3.750 ns, 0 failing of 75 endpoints. Fmax estimate = 1 / (10 ns - 0.354 ns) = 1 / 9.646 ns, about 103.7 MHz. This is an estimate from a single run at one constraint. Confirming the true limit would need reruns with a tighter clock.

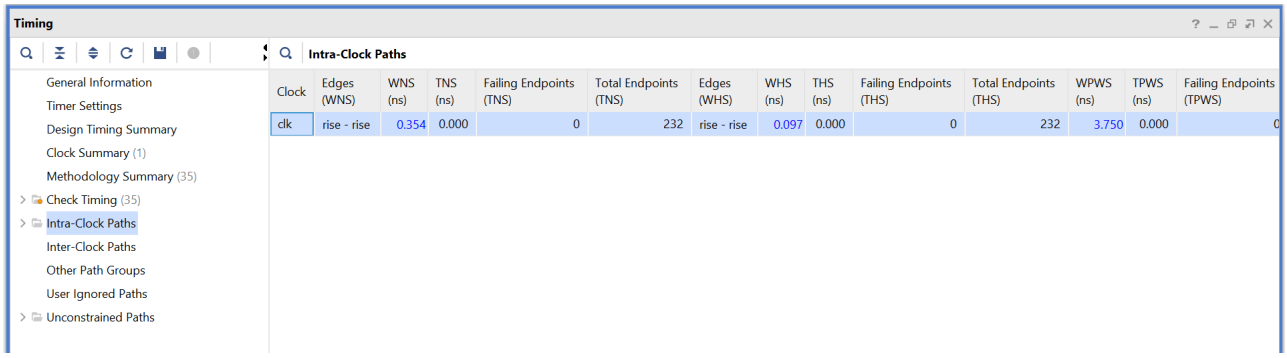


Figure 10. *Intra-clock paths table after implementation.*

> **In plain terms.** The design has a single clock, so this table has a single row. It lists paths that start and end on the same clock and repeats the summary figures: all slack values are positive and nothing fails.

Technical detail. Clock `clk`, rise-to-rise edges: WNS 0.354 ns, TNS 0.000 ns, 0 failing of 232 setup endpoints; WHS 0.097 ns, 0 failing of 232 hold endpoints; WPWS 3.750 ns. An endpoint is a point where a signal path ends and is checked against its deadline.

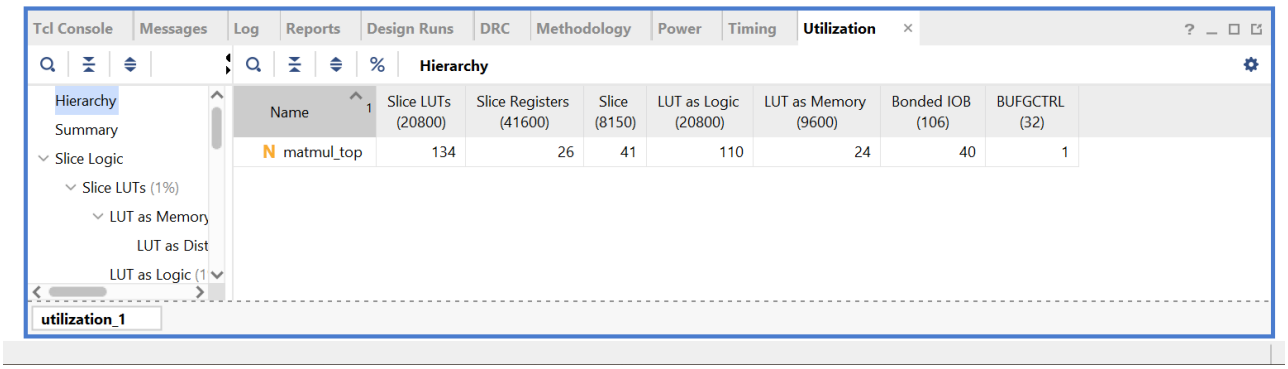


Figure 11. *Post-implementation utilization.*

> **In plain terms.** After placement and routing, the design needs slightly fewer building blocks than after synthesis: 134 logic elements, of which 110 do logic and 24 act as small memories. The 26 registers are unchanged.

Technical detail. Slice LUTs 134 / 20,800 (0.64%); slice registers 26 / 41,600 (0.06%); slices 41 / 8,150 (0.50%); LUT as logic 110 / 20,800 (0.53%); LUT as memory 24 / 9,600 (0.25%); bonded IOB 40 / 106; BUFGCTRL 1 / 32. The drop from 140 to 134 LUTs is consistent with further optimization during implementation.

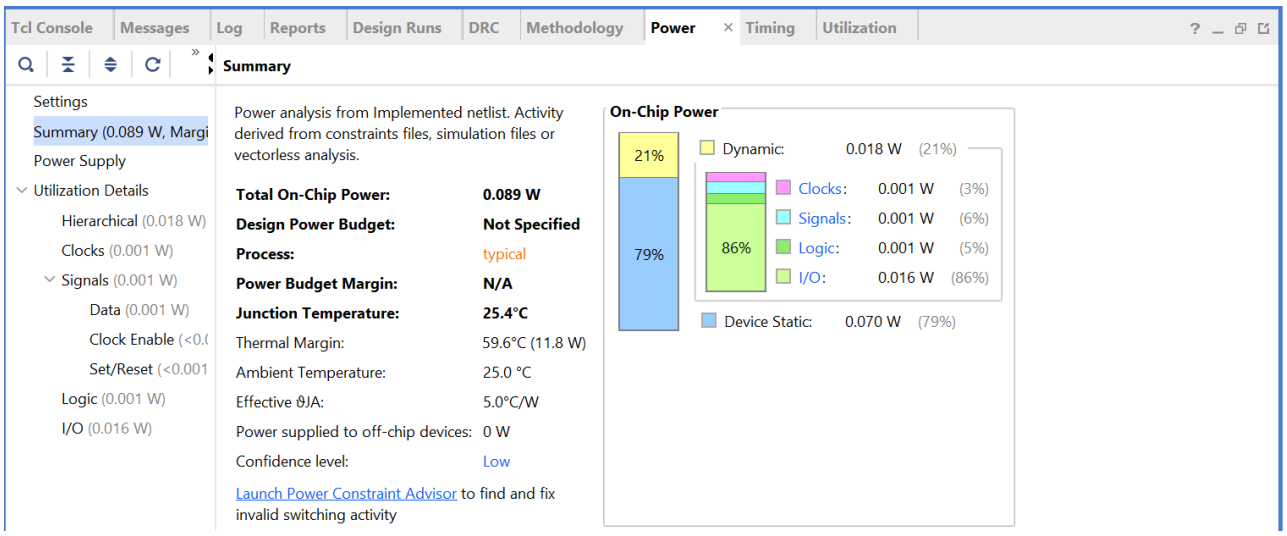
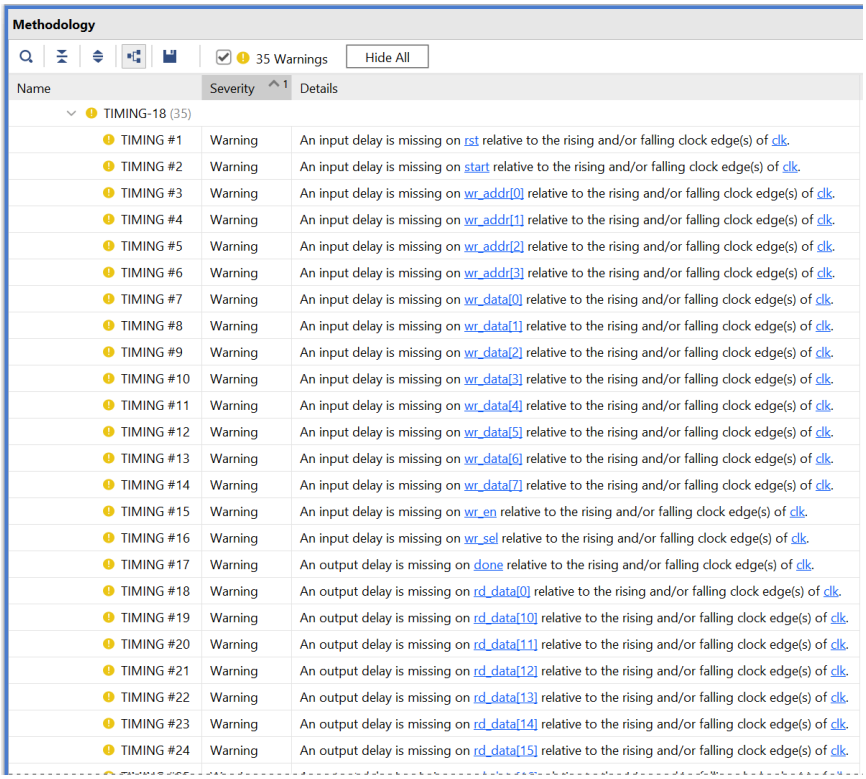


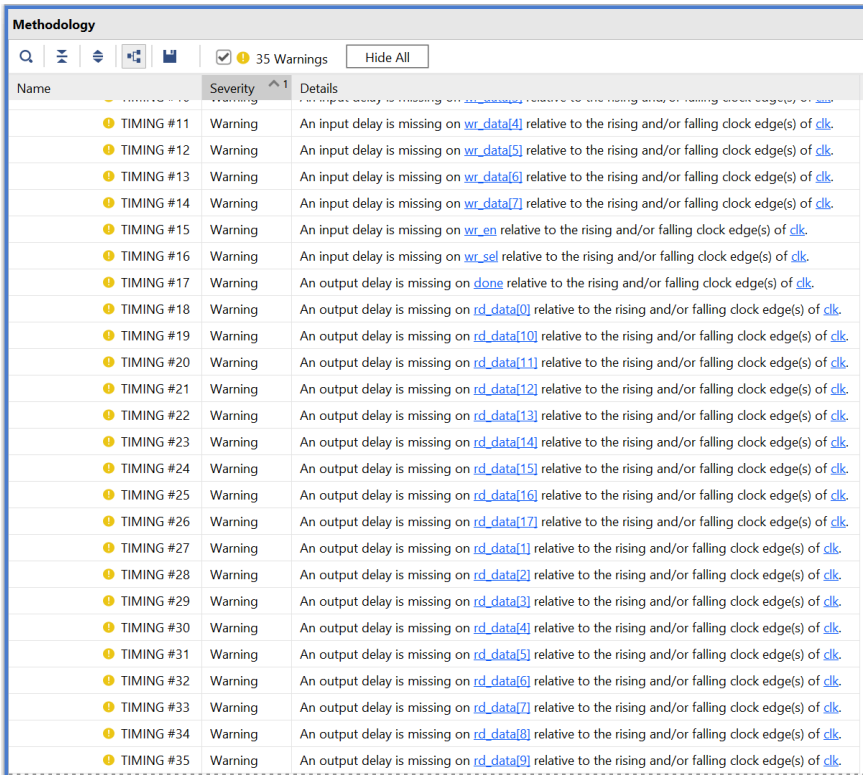
Figure 12. *Post-implementation power report.*

In plain terms. A chip uses electricity in two ways. Static power is consumed simply by being switched on, like a standby light. Dynamic power is the extra used when signals switch. The estimate is 0.089 W in total, and about four fifths of it is static. The small dynamic part is mostly the 40 input and output pins. The estimate is rough: Vivado rates its own confidence as low, because it did not have real activity data from the running design.

Technical detail. Total on-chip power 0.089 W: device static 0.070 W (79%) and dynamic 0.018 W (21%). The dynamic share splits into clocks 0.001 W (3%), signals 0.001 W (6%), logic 0.001 W (5%) and I/O 0.016 W (86%). Junction temperature 25.4 °C, thermal margin 59.6 °C (11.8 W), ambient 25.0 °C, effective θJA 5.0 °C/W, process "typical", no power budget specified. Confidence level: low, because the analysis is vectorless (default switching assumptions instead of simulation activity).



(a) Top of the list



(b) Scrolled to the end of the list

Figure 13. *Vivado methodology report: 35 warnings, all of type TIMING-18.*

> **In plain terms.** Vivado's methodology report is a quality checklist. It raised 35 warnings, all of the same kind: the 40-pin interface has no timing assumptions for signals coming from or going to the outside world ("input delay missing", "output delay missing"). With no circuit board in this project there is nothing real to describe, so these warnings are expected. They mean that the timing results above cover paths inside the chip only.

Technical detail. 35 warnings, all TIMING-18: 16 inputs (`rst`, `start`, `wr_en`, `wr_sel`, `wr_addr[3:0]`, `wr_data[7:0]`) and 19 outputs (`done`, `rd_data[17:0]`). The clock is constrained, while input and output delays were deliberately left unconstrained, so the reported slack covers internal register-to-register paths only. `rd_addr` is not listed because it feeds the outputs through combinational logic only.

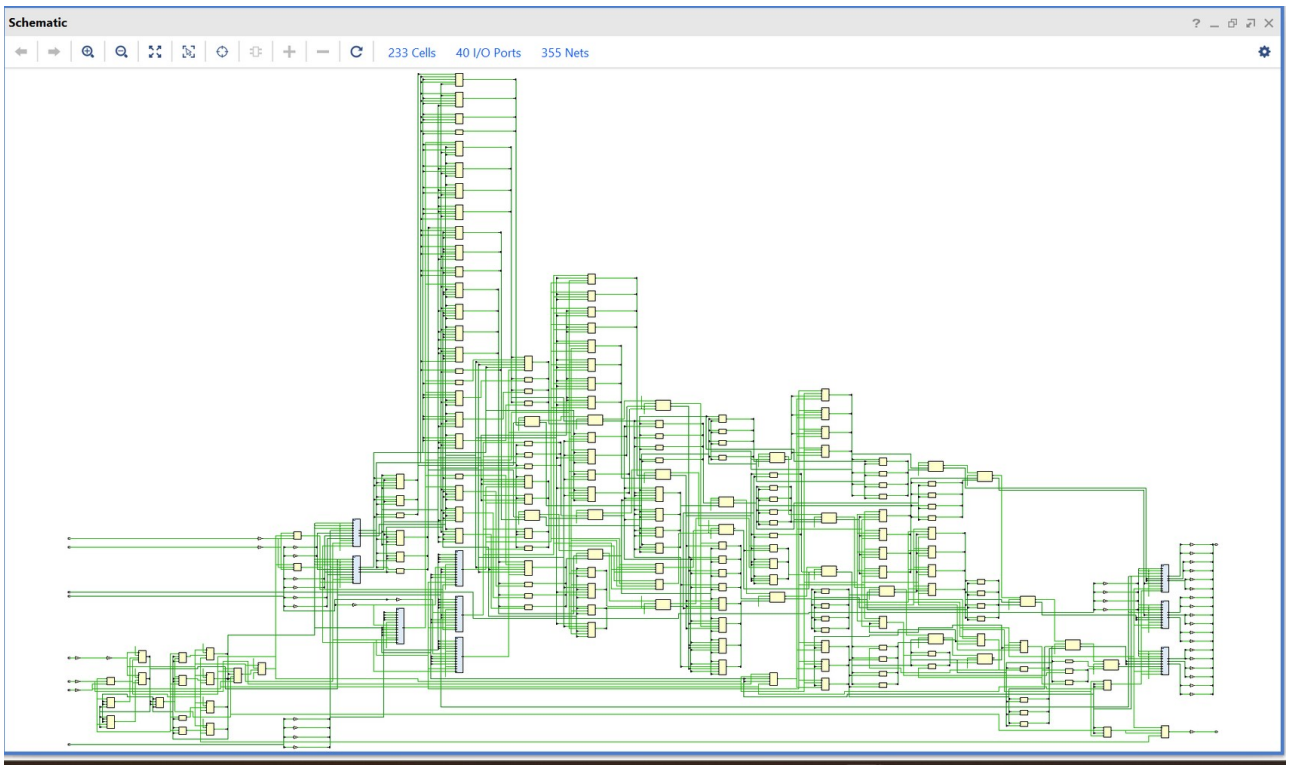


Figure 14. Schematic of the implemented design (233 cells, 40 I/O ports, 355 nets).

> **In plain terms.** This is the same circuit after it has been mapped onto the FPGA's real building blocks. It looks much busier than Figure 1 because the generic blocks have been broken down into the chip's small primitive elements. It is included for completeness and is not meant to be read gate by gate.

Technical detail. Post-implementation netlist: 233 cells, 40 I/O ports, 355 nets (compared with 71 cells and 162 nets in the RTL schematic). Cells are primitives such as LUTs, flip-flops, I/O buffers and the clock buffer.

4.4 Synthesis versus implementation

Metric	After synthesis	After implementation	Change
Slice LUTs	140	134	-6
Slice registers	26	26	none
Setup slack (WNS)	+0.379 ns	+0.354 ns	-0.025 ns
Hold slack (WHS)	+0.098 ns	+0.097 ns	-0.001 ns
Failing endpoints	0	0	none
Estimated Fmax	about 104 MHz (1 / 9.621 ns)	about 104 MHz (1 / 9.646 ns)	negligible

5. Scope and limitations

The results are reported as measured, together with their limits:

- **Tool estimates, not hardware measurements.** No bitstream was generated and nothing was run on a physical FPGA board. All FPGA figures come from Vivado reports.
- **Simulation is functional.** Verification used behavioral simulation. No post-synthesis or post-implementation timing simulation was run.

- **Not exhaustive.** 205 tests include extreme values and 200 random pairs, but the full input space is far too large to cover completely, and no formal coverage metrics were collected.
- **Fmax is an estimate.** It is derived from the slack of one run at a 10 ns constraint, not from a search for the fastest passing clock.
- **I/O timing is unconstrained.** Slack figures cover internal paths only (see Figure 13).
- **Power is a low-confidence estimate.** It is vectorless and should not be compared across platforms.
- **Platform-specific resource mapping.** The memories were mapped to LUT RAM on this FPGA. Other targets may use registers or block RAM, so area figures do not transfer directly.
- **Educational scope.** A single shared multiplier was implemented. A fully parallel version for a speed-versus-area comparison was not built.

6. Planned next step

The same design is planned to be taken through an open-source RTL-to-GDSII flow (OpenLane with the SkyWater 130 nm process) to compare FPGA and ASIC results for area, timing and power. No ASIC results are included in this report.